

Not All Tokens Are Equal: Uneven Uncertainty and Structure in Language Generation

Abstract

Large Language Models (LLMs) are typically viewed as uniform stochastic systems, where every token requires full neural computation. We challenge this assumption by showing that generation is fundamentally heterogeneous, consisting of near-deterministic grammatical constraints and high-entropy semantic decisions.

We model generation as a transition system over part-of-speech contexts and extract deterministic finite automata (DFA) that capture these dynamics. Our analysis reveals low-entropy regions within the state space with predictable transitions and high-entropy regions associated with ambiguity and elevated error rates. Flow analysis further shows that natural language trajectories remain within stable regions, while random transitions rapidly collapse into absorbing states.

Building on this, we evaluate a simple hybrid inference strategy that routes low-entropy states to a lightweight n-gram model and high-entropy states to a neural model, finding that up to 25% of tokens can bypass expensive inference with minimal performance loss.

These results suggest that LLM computation can be adaptively allocated based on uncertainty, enabling more efficient and interpretable generation.

1. Introduction

Large Language Models (LLMs) are typically treated as monolithic stochastic systems, where each token is generated through a full forward pass through billions of parameters. This uniform treatment assumes that all tokens require equivalent computational effort, despite the highly structured nature of natural language.

In this work, we challenge this assumption. We posit that LLM generation is fundamentally heterogeneous, consisting of two distinct regimes: a high-entropy semantic regime, where multiple continuations are plausible and full neural inference is required, and a low-entropy structural regime, governed by near-deterministic grammatical constraints. Crucially, these regimes are not rare edge cases but coexist throughout generation, suggesting that a significant portion of token decisions may not require full model capacity.

Our central hypothesis is that a substantial fraction of next-token predictions are determined by low-entropy, state-dependent transitions that can be captured without full neural computation. By viewing the model as a transition system over Part-of-Speech (POS) contexts, we demonstrate that it is possible to extract deterministic finite automata (DFAs) that encode these structural dynamics. These DFAs reveal a compact representation of a "grammatical manifold" that governs large portions of the generation process.

We empirically validate this perspective through three key findings. First, we show that entropy provides a reliable signal of grammatical uncertainty: low-entropy regions correspond to near-deterministic transitions, while high-entropy regions correspond to intrinsically ambiguous contexts. Second, we analyze traversal dynamics over the learned DFAs and find that while random walks rapidly collapse into absorbing configurations, natural language trajectories remain within stable regions and only transiently enter these sinks before immediately recovering. Third, we evaluate a simple hybrid inference strategy that routes low-entropy states to a lightweight n-gram model and a neural model. We show that up to 25% of tokens can be handled by the cheaper model with minimal degradation in performance, providing direct evidence that not all tokens require equal computational effort.

The remainder of the paper is organized as follows. Section 2 reviews related work on finite state models and automata extraction. Section 3 formalizes the problem and introduces the context representations used in this study. Section 4 describes the experimental setup and data generation process. Section 5 outlines the DFA construction and analysis methods. Section 6 presents empirical results on structure, uncertainty, and dynamics. Section 7 discusses implications and directions for future work.

2. Background and Related Work

2.1 Finite State Machines

Finite state machines (FSMs) provide a formal framework for modeling sequential processes with a finite number of discrete states. An FSM consists of a set of states, a finite input alphabet, and a transition function that governs how the system evolves in response to input symbols. At each step, the machine transitions from its current state to a new state based on observed input, producing a sequence of state transitions that capture the system's behavior over time.

In this work, we focus specifically on deterministic finite automata. A DFA is a type of FSM defined as a tuple $(Q, \Sigma, \delta, q, F)$, where Q is a finite set of states, Σ is the input alphabet, δ is a deterministic transition function, q is the initial state, and F is a set of accepting states. Determinism ensures that for each state and input symbol, there is exactly one next state, yielding a unique state trajectory for any input sequence.

While DFAs are traditionally used to recognize formal languages, they can also be used to approximate the behavior of complex sequential models. We use DFAs as a tool to analyze the dynamics of

grammatical structure during autoregressive text generation. In our setting, transitions are induced by model-generated continuations rather than externally defined inputs. States correspond to part-of-speech (POS) context representations, and transitions are defined by appending the next predicted POS tag. Accepting states correspond to contexts that support a target grammatical class. This formulation enables direct analysis of determinism, ambiguity, and structural stability in LLM-generated text.

2.2 Automata Extraction from Neural Models

The abstraction of finite state automata from neural networks is an area that has been well studied. Early approaches focused on recurrent neural networks (RNNs), showing that their hidden states can implicitly encode regular language structure [4]. Methods such as state clustering and L^* active learning have been used to approximate deterministic finite automata that replicate the behavior of trained RNNs on formal language tasks [10, 12, 11, 13, 9]. State clustering operates by grouping continuous hidden states into discrete equivalence classes, enabling the construction of approximate automata. In contrast, the L^* algorithm learns a minimal DFA by issuing a sequence of membership and equivalence queries to a teacher (oracle) model [5].

More recent work has extended these ideas to transformer-based architectures. Adriaensen and Maene [6] and Zhang et al. [14] independently applied the L^* algorithm to extract DFAs from encoder-only transformer models. In another study hidden-layer embeddings from an LLM were clustered into discrete states, enabling the detection of hallucination patterns through analysis of induced transition dynamics [15].

Despite these contributions, existing automata extraction methods exhibit several limitations. First, they are typically evaluated on formal or synthetic languages, where the underlying structure is well-defined and relatively simple [7, 13, 14, 6]. Second, clustering-based approaches attempt to map continuous representations into discrete states, which can introduce ambiguity and reduce interpretability. Third, and most importantly, most methods generally focus on modeling whether a sequence belongs to a language rather than capturing the dynamics of sequence generations.

In contrast, our approach constructs DFAs directly from observable sequence behavior rather than from internal model representations. This avoids the need for clustering high-dimensional activations and yields a DFA with explicit, interpretable states grounded in linguistic structure. By modeling transitions over autoregressive outputs, our method captures the temporal evolution of grammatical structure.

3. Method

3.1 Problem Formulation

We study the dynamics of grammatical structure in autoregressive text generation by modeling transitions over part-of-speech (POS) sequences. Let $x = (w_1, w_2, \dots, w_T)$ denote a sequence of tokens generated by a language model, and let $p = (p_1, p_2, \dots, p_T)$ denote the corresponding sequence of POS tags obtained via an external tagger. We consider the evolution of grammatical structure as a sequence

of transitions over POS contexts.

At each time step t , we define a fixed-length POS context:

$$s_t = (p_{t-k}, \dots, p_{t-1})$$

where k is the context window size. This representation is used consistently across all experiments and serves as input to the learned classifiers.

Given a target POS class $c \in \Sigma$, where Σ is the set of POS tags, we define a binary classification task:

$$y_t = \Delta[p_t = c]$$

where $y_t = 1$ if the next POS tag matches the target class and $y_t = 0$ otherwise. The goal is to learn a function $f: s_t \rightarrow [0, 1]$ that estimates the probability that a given context leads to the target POS class.

We train neural sequence classifiers to approximate this function using observed POS transitions from LLM-generated text. These classifiers serve as oracles that define the acceptance behavior of states in a finite-state representation. Specifically, each context s_t is treated as a state, and the classifier's prediction $f(s_t)$ determines whether the state is considered accepting under a chosen threshold.

Using this formulation, we model grammatical structure as a state transition system over POS contexts. Transitions are defined by appending the next POS tag and updating the context:

$$\delta(s_t, p_t) = (p_{t-k+1}, \dots, p_t)$$

This induces a deterministic transition function over the state space, allowing us to construct a DFA that approximates the behavior of the learned classifier.

While the classifier always operates on fixed-length POS contexts, the distribution of next POS tags p_t depends on how the underlying language model is prompted during generation. This distinction is formalized in the following section.

3.2 Context Definitions

The behavior of grammatical transitions from the LLM depends not only on the POS context used for classification but also on how the language model is conditioned when generating the next token. In this work, we consider two prompting regimes: local (k-gram) context and prefix-conditioned context.

Local (k-gram) Context

In the local setting, the language model is prompted with a fixed-length window of the previous k tokens:

$$(w_{t-k}, \dots, w_{t-1})$$

The model then generates the next token w_t , which is subsequently mapped to a POS tag p_t . This setup induces a locally conditioned transition distribution, where the next POS depends only on a bounded token history.

Importantly, the classifier does not operate on tokens but on the corresponding POS sequence:

$$s_t = (p_{t-k}, \dots, p_{t-1})$$

This formulation approximates a Markovian process over POS tags, capturing short-range grammatical dependencies such as determiner–noun agreement and auxiliary–verb constructions.

Prefix-Conditioned Context

In the prefix-conditioned setting, the language model is prompted with the full token prefix:

$$(w_1, \dots, w_{t-1})$$

The model generates the next token w_t , which is again mapped to a POS tag p_t . This induces a globally conditioned transition distribution, where the next POS may depend on long-range context, including clause structure and discourse-level dependencies.

In our implementation, the prefix is defined at the sentence level. For a given position t , the prefix includes all tokens from the beginning of the current sentence up to position $t-1$. If the context window spans a sentence boundary, the prefix additionally includes the immediately preceding sentence. This definition provides bounded long-range context while avoiding unbounded document-level conditioning.

Despite this difference in conditioning, the classifier input remains unchanged:

$$s_t = (p_{t-k}, \dots, p_{t-1})$$

Thus, the distinction between local and prefix-conditioned contexts lies entirely in how the next token is generated, not in how the classifier processes the input.

	Local Context	Prefix-Conditioned
Input Tokens	to be enrolled, and	Upon which the youth of Ava , and the neighbouring places , came eagerly to be enrolled , and
Input POS Sequence	PART AUX VERB PUNCT CCONJ	SCONJ DET DET NOUN ADP PROPN PUNCT CCONJ DET VERB NOUN PUNCT VERB ADV PART AUX VERB PUNCT CCONJ
LLM Next Token	the	were
Next Token POS Tag	DET	AUX
Classifier Input	PART AUX VERB PUNCT CCONJ	PART AUX VERB PUNCT CCONJ

Figure 1: Local vs. prefix-conditioned prompting. The classifier input (POS k-gram) is identical in both settings, while the language model is conditioned either on a fixed local window or the full sentence prefix.

Comparison and Role in This Work

These two prompting regimes provide complementary views of grammatical structure. By comparing transition behavior under these two regimes while holding the state representation fixed, we can analyze:

- the extent to which grammatical behavior is locally determined,
- how long-range context alters transition distributions, and
- where the two formulations agree or diverge.

This comparison plays a central role in our analysis, enabling us to identify regions of stable grammatical structure as well as areas of intrinsic ambiguity in LLM-generated text.

3.3 DFA Construction

We construct DFAs to model grammatical transitions over POS contexts. Each state corresponds to a fixed-length POS sequence $s_t = (p_{t-k}, \dots, p_{t-1})$, and transitions are defined by appending the next POS tag and shifting the context window:

$$\delta(s_t, p_t) = (p_{t-k+1}, \dots, p_{t-1})$$

In principle, the full state space consists of all possible POS k-grams, which grows exponentially with k. Rather than enumerating this full space, we construct the DFA over the observed manifold of contexts present in the data. We include only those states that appear in the dataset, along with transitions observed between them.

Transitions that lead to unseen contexts are mapped to a designated sink state, which captures out-of-distribution or unobserved grammatical configurations. This yields a finite and tractable automaton that reflects the empirical structure of POS transitions induced by the language model.

For a given target POS class c , each state s_t is assigned an acceptance value based on the classifier prediction $f(s_t)$. A state is considered accepting if $f(s_t) \geq \tau$, where τ is a fixed threshold. This defines a target-specific DFA that captures contexts likely to produce the desired grammatical class.

To reveal the underlying structure of the transition system, we apply DFA minimization using Hopcroft's algorithm. Two states are considered equivalent if they are indistinguishable with respect to future acceptance behavior under all possible input sequences. Minimization merges such states into equivalence classes while preserving the language recognized by the automaton. This step significantly reduces the size of the DFA while providing a more interpretable representation of grammatical dynamics.

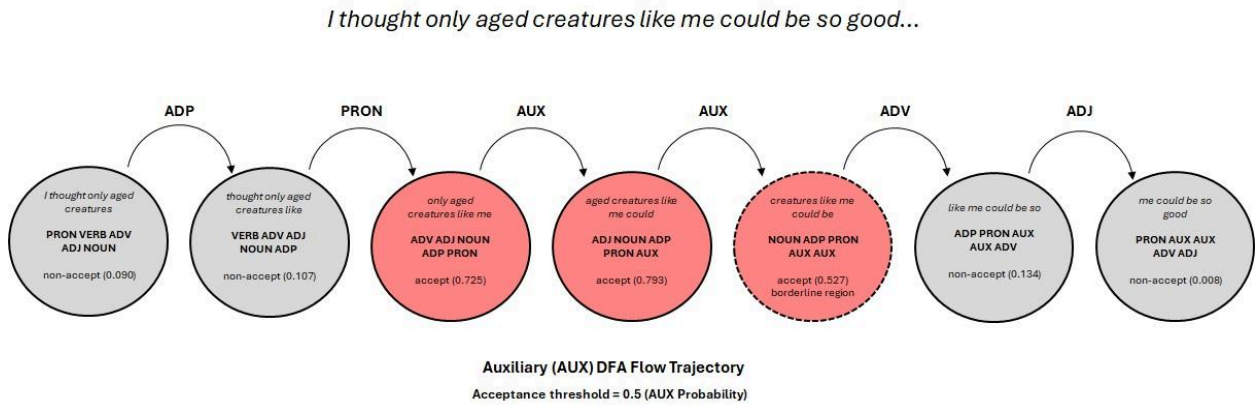


Figure 2: Example traversal through the AUX prefix-conditioned DFA using a naturally occurring sentence. Each node represents a POS 5-gram state and is labeled with the classifier’s predicted probability for AUX continuation.

4. Experimental Setup

4.1 Data Sources

We construct datasets of POS transitions from text generated by a pretrained autoregressive language model. All experiments use GPT-2 (124M parameters) to generate token sequences under the two prompting regimes described in Section 3.2. Unless otherwise specified, sequences are generated using greedy decoding (temperature = 0). This allows us to isolate structural properties independent of sampling variability. The effects of stochastic decoding are explored separately in Sections 5 and 6.

To ensure diversity in grammatical structure, we draw prompts from two sources:

- **Project Gutenberg:** We sample 995 English texts from Project Gutenberg, a collection of over 75,000 publicly available eBooks. The selected texts span a range of genres and writing styles, with publication dates between 1993 and 2026.
- **Dialog from meeting transcripts:** We collect conversational data from meeting transcripts across Kwaai's online working groups. These texts consist of informal, spoken-style language with shorter, less structured utterances.

All source documents are processed using the POS tagger provided by spaCy's NLP pipeline. From these corpora, we sample approximately one million sequences from the Gutenberg dataset and approximately 1.5 million sequences from the dialog dataset.

4.2 POS Tagging and Preprocessing of Generated Tokens

For each prompt, we generate continuations from GPT-2 under both local and prefix-conditioned regimes. Each generated token is then mapped to a POS tag using spaCy.

To improve tagging reliability, we generate $k - 1$ additional tokens beyond the immediate next token. The POS tagger is applied to the concatenated sequence consisting of the prompt, the next generated token, and the additional continuation. This provides sufficient surrounding context for accurate POS assignment, mitigating errors arising from limited local context.

We apply the following preprocessing steps:

- **Whitespace filtering:** Sequences where the generated continuation consists solely of whitespace are excluded from neural model training, as these do not provide meaningful grammatical information.
- **Tokenization alignment:** Tokens that cannot be reliably mapped to POS tags due to misalignment between GPT-2's subword tokenization and spaCy's word-level tokenization are excluded from neural model training. These cases account for approximately 1–2% of the dataset.

These preprocessing steps ensure that the resulting POS sequences accurately reflect grammatical structure while minimizing noise from tokenization artifacts.

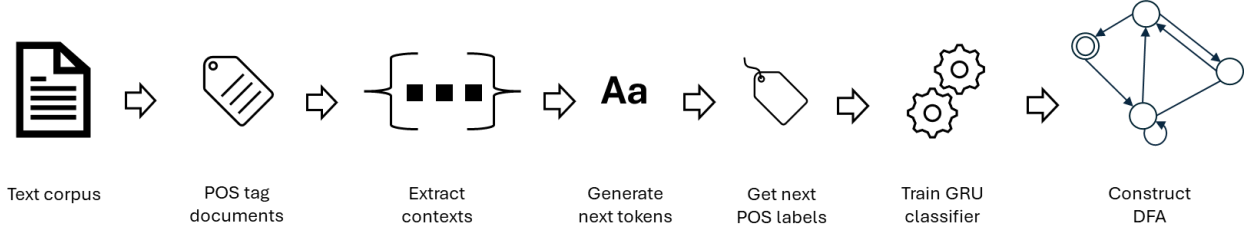


Figure 3: DFA construction pipeline. Source documents are first POS-tagged, and both fixed-length (local) and prefix-conditioned contexts are extracted from the tagged text. These contexts are used to prompt the LLM for next-token generation, and the POS labels of the generated continuations form context–label pairs. These pairs are used to train GRU classifiers for target POS classes. Classifier predictions, together with deterministic k -gram transitions, define the DFA, which is subsequently minimized for structural analysis.

5. Experiments

5.1 Overview

We conduct a series of experiments to analyze the structure and dynamics of grammatical transitions induced by LLM-generated text. Our evaluation is designed to compare behavior across context regimes, quantify uncertainty in POS transitions, and characterize the structural properties of the finite-state representations.

DFA construction in the experiments is performed on a subset of target POS classes selected for stability of transition behavior. The full set of targets and selection criteria are described in Section 6.

To assess the robustness of grammatical structure across domains, all experiments are performed on both the Gutenberg and Dialog datasets described in section 4.1.

5.2 Baseline Comparison

To evaluate whether neural models capture structure beyond simple frequency patterns, we compare multi-class GRU-based classifiers against a frequency-based baseline.

For each POS context s_t , the baseline predicts the most frequently observed next POS tag:

$$\hat{p}_t = \operatorname{argmax}_{c \in \Sigma} \operatorname{count}(s_t, c)$$

If a context is unseen, the model defaults to the most frequent POS tag in the dataset. We evaluate both models using classification accuracy and macro-F1 score across all target POS classes and datasets.

5.3 Context Regime Comparison

To assess the effect of long-range context, we construct DFAs under both local and prefix-conditioned prompting regimes. For each POS class, we compare the acceptance behavior of corresponding states across the two DFAs.

We compute:

- **State-level agreement rate**, defined as the proportion of states for which both DFAs assign the same acceptance label
- **Cohen's kappa**, to measure agreement beyond chance

These metrics are computed over the shared set of observed POS contexts.

5.4 Entropy-Based Analysis

To quantify uncertainty in grammatical transitions, we compute the entropy of the next-POS distribution for each state:

$$H(s_t) = - \sum_{c \in \Sigma} P(c|s_t) \log P(c|s_t)$$

Where Σ is a set of POS tags and $P(c|s_t)$ is estimated using empirical transition frequencies.

We compute entropy separately under both context regimes, yielding a local entropy $H_{local}(s_t)$ and prefix-conditioned entropy $H_{prefix}(s_t)$. To obtain a unified measure of uncertainty for each state, we define mean entropy as

$$H_{mean}(s_t) = \frac{H_{local}(s_t) + H_{prefix}(s_t)}{2}$$

This formulation captures uncertainty under both conditioning strategies while maintaining a single scalar measure for analysis. For aggregate statistics, we report the average of $H_{mean}(s_t)$ within each group of states.

To relate uncertainty to model behavior, analyze its relationship to classification performance in two complementary ways. First, we associate each state with classification outcomes (e.g., true positive, false positive, false negative) derived from model predictions on validation data. Second, we examine the relationship between entropy and classification error rate as a continuous function over the state space.

To analyze how uncertainty relates to structural consistency, we group states into entropy buckets based on $H_{mean}(s_t)$.

For each bucket, we compute:

- mean entropy
- state-level agreement rate between local and prefix-conditioned DFAs
- Cohen's kappa within each bucket

5.5 DFA Structural Analysis

We analyze the structural properties of the constructed DFAs using graph-based metrics that characterize how states and acceptance behavior are distributed and how grammatical structure is organized.

We focus on measures of structural concentration, including

- the fraction of states contained within the largest strongly connected component (SCC)
- the fraction of accepting state within the largest SCC
- the relative density of accepting states within local neighborhoods of the state space

We additionally examine the distribution of accept scores across states and across various SCC sizes.

5.6 Flow Dynamics

To study the evolution of grammatical transitions, we perform traversal experiments over the DFA. Starting from selected initial states, we simulate sequences of transitions by iteratively applying the transition function.

For each traversal, we record:

- number of steps before reaching recurrent or stable regions of the state space
- number of unique states visited
- frequency of visits to accepting states

We perform experiments under multiple initialization strategies, including random states and states aligned with the target POS class.

To characterize how transitions distribute over the state space, we compute measures of flow concentration and persistence, including:

- visitation entropy and Gini coefficients over state visitation frequencies
- the proportion of visitation mass concentrated within the largest SCC
- the rate at which trajectories remain within accepting regions once entered

In addition, we analyze the structural properties of states with disproportionately high incoming transition volume. For these states, we compute:

- distribution of predecessor states and their SCC memberships
- frequency of incoming transitions from different regions of the state space

- POS symbols most commonly associated with transitions into these states

5.7 Stability Under Stochastic Decoding

To evaluate the robustness of grammatical transition structure decoding variability, we perform a temperature sweep over generation parameters using the Gutenberg dataset.

We construct a stratified set of POS contexts based on their empirical frequency in the dataset. We select:

- all high-frequency POS contexts with more than 100 occurrences
- 2,000 contexts sampled from medium-frequency groups (more than 10 occurrences)
- 2,000 contexts sampled from low-frequency groups (more than 2 occurrences)
- 1,000 contexts with a single occurrence

For each selected POS context, we sample a subset of raw sequences corresponding to that context, with the number of samples set to 10% of its observed frequency. This ensures proportional representation across frequency levels while controlling computational cost.

For each sampled sequence, we generate 10 continuations using GPT-2 at temperature values $T \in \{0.0, 0.2, 0.5, 0.8\}$ using the prefix-conditioned prompting strategy described in Section 3.2. Each generated token is mapped to a POS tag using the procedure described in Section 4.

To quantify differences in transition behavior, we compute the Jensen-Shannon (JS) divergence between the next-POS distributions obtained at ($T = 0$) and those obtained at higher temperatures for each context.

To analyze structural effects, we construct empirical temperature-specific DFAs from the generated transitions at each temperature setting. For each temperature, we aggregate the observed next-POS outcomes for each POS context and define transitions using the same fixed k-gram update rule described in Section 3. Acceptance is then assigned empirically for each target POS class based on the observed next-POS distribution at that temperature by labeling a state as accepting when the empirical probability of the target class exceeds the analysis threshold. This yields a temperature-conditioned DFA derived directly from observed transition behavior rather than from classifier predictions.

We then minimize these empirical DFAs and compare their structural and flow-based properties using the metrics described in Sections 5.5 and 5.6.

6. Results

6.1 GRU Achieves Consistent Improvements Over Baseline

Across both the Gutenberg and dialog datasets, GRU models achieve modest but consistent improvements over the baseline in overall accuracy and macro-F1 score under both local and prefix-conditioned regimes. These gains are more pronounced in the local setting and tend to diminish under prefix conditioning, where both models exhibit reduced performance due to increased variability in the next-POS distribution.

Performance varies substantially across POS classes. Categories such as auxiliary (AUX), noun (NOUN), and determiner (DET) achieve relatively strong recall and F1 scores across both datasets and prompting strategies, indicating stable and predictable transition behavior. For example, AUX and DET consistently exhibit recall and F1 scores in the mid-to-high range, while NOUN maintains strong recall under both local and prefix-conditioned settings. In contrast, categories such as adverb (ADV), subordinating conjunction (SCONJ), and particle (PART) show significantly lower recall and F1 scores, reflecting higher variability and weaker predictive structure under our modeling scheme.

The relative performance of the GRU and baseline models provides insight into the nature of grammatical dependencies. For highly frequent and locally predictable classes (e.g., DET, NOUN), the baseline performs competitively, suggesting that much of their structure can be explained by local co-occurrence statistics. However, for context-sensitive categories such as AUX and coordinating conjunction (CCONJ), the GRU exhibits clearer advantages, indicating that these transitions depend on patterns not captured by simple frequency counts.

Based on these results, we restrict DFA construction to a subset of stable target classes, defined as those exhibiting moderate to strong recall and F1 scores across both local and prefix-conditioned regimes. This includes categories such as AUX, NOUN, and DET, which provide sufficient predictive reliability for meaningful structural analysis.

We additionally evaluate CCONJ as a contrast case. Although its performance is less stable in the Gutenberg corpus, it exhibits stronger and more consistent behavior in the dialog dataset. This behavior likely reflects the conversational nature of the data, which contains a higher density of clause-linking structures that make coordinating conjunctions more predictable. We therefore include CCONJ to contrast with more stable targets, evaluating it exclusively on the dialog dataset.

Categories with consistently low recall or highly variable performance are excluded, as their transition behavior does not support stable finite-state representations.

For each selected target, we train binary GRU classifiers, construct the corresponding DFAs, and use these models for the analyses described in Section 5.

			GUTENBERG				DIALOG			
			LOCAL		PREFIX		LOCAL		PREFIX	
	POS	MODEL	RECALL	F1	RECALL	F1	RECALL	F1	RECALL	F1
HIGH-STABILITY TARGETS	AUX	Baseline	0.564	0.559	0.499	0.509	0.643	0.658	0.596	0.603
		GRU	0.584	0.576	0.537	0.537	0.666	0.679	0.624	0.618
	NOUN	Baseline	0.650	0.614	0.678	0.628	0.609	0.589	0.612	0.581
		GRU	0.694	0.626	0.740	0.640	0.643	0.607	0.661	0.614
	DET	Baseline	0.669	0.641	0.589	0.566	0.674	0.594	0.596	0.539
		GRU	0.695	0.646	0.598	0.580	0.685	0.593	0.636	0.561
CONTEXT-DEPENDENT TARGET	CCONJ	Baseline	0.594	0.541	0.505	0.433	0.775	0.721	0.601	0.568
		GRU	0.701	0.568	0.587	0.480	0.779	0.734	0.648	0.588
LOW-STABILITY TARGETS	PRON	Baseline	0.290	0.361	0.299	0.344	0.409	0.468	0.474	0.494
		GRU	0.306	0.387	0.319	0.362	0.441	0.489	0.507	0.531
	ADP	Baseline	0.440	0.416	0.435	0.411	0.375	0.383	0.342	0.349
		GRU	0.516	0.458	0.497	0.440	0.405	0.401	0.358	0.359

Table 1: Baseline vs. GRU classification performance across datasets and prompting regimes. AUX, NOUN, and DET are treated as stable targets due to their consistently strong performance. CCONJ exhibits context-dependent behavior, performing well in dialog but less consistently in Gutenberg. PRON and ADP represent unstable targets with lower and more variable performance.

6.2 Consistent Structure Across Context Regimes

We observe consistently high agreement in the DFAs constructed under local and prefix-conditioned prompting across both the Gutenberg and dialog datasets. For AUX, NOUN, and DET, overall state-level agreement exceeds 90% in all settings, with agreement reaching as high as 95.2% for NOUN in the Gutenberg dataset (Table 2). Corresponding Cohen’s kappa scores range from approximately 0.79 to 0.87, indicating strong agreement beyond chance and confirming that the induced structure is largely consistent across conditioning strategies.

This pattern holds across datasets, although agreement is slightly lower in the dialog dataset, reflecting increased variability in conversational language. AUX, for example, achieves an agreement rate of 94.3% ($\kappa = 0.846$) in the Gutenberg dataset compared to 93.1% ($\kappa = 0.805$) in dialog, while DET shows a similar reduction from 93.9% ($\kappa = 0.840$) to 91.9% ($\kappa = 0.795$). Despite these differences, the overall high level of agreement remains high. This suggests that a substantial portion of grammatical structure is locally determined and robust to broader context.

The majority of states fall into the reject-both and accept-both categories, indicating strong alignment between the two models across most of the state space. Disagreement is limited to a relatively small fraction of states, typically accounting for less than 6–8% of the total, and is distributed across both local-only and prefix-only acceptance cases.

In contrast, CCONJ exhibits a different pattern. While overall agreement remains high in the dialog dataset (93.8%), the Cohen’s kappa score is substantially lower ($\kappa = 0.73$). Additionally, the prefix-conditioned model produces a noticeably higher acceptance rate (15.8%) compared to the local model (10.3%), suggesting that coordinating conjunction transitions are more sensitive to broader context. This aligns with the role of conjunctions in linking clauses and maintaining discourse coherence

which may not be fully captured within a fixed local window.

Taken together, these results indicate that grammatical structure in LLM-generated text consists of a stable, locally determined core that is supplemented by context-sensitive regions where long-range dependencies influence transitions.

POS	DATASET	AGREEMENT %	KAPPA	ACCEPT BOTH %	REJECT BOTH %	LOCAL ONLY %	PREFIX ONLY %	LOCAL ACCEPT %	PREFIX ACCEPT %
AUX	Gutenberg	94.286	0.846	21.710	72.576	2.900	2.814	24.610	24.524
	Dialog	93.066	0.805	19.674	73.391	3.301	3.633	22.975	23.308
NOUN	Gutenberg	95.186	0.868	21.462	73.724	1.811	3.003	23.273	24.466
	Dialog	93.534	0.818	19.807	73.727	2.975	3.491	22.782	23.298
DET	Gutenberg	93.913	0.840	22.524	71.389	3.415	2.673	25.938	25.196
	Dialog	91.852	0.795	23.195	68.658	5.010	3.137	28.205	26.332
CCONJ	Dialog	93.850	0.730	9.953	83.896	0.338	5.812	10.291	15.766

Table 2: Agreement between local and prefix-conditioned DFAs, reported as percentages of state-level outcomes. Agreement is decomposed into accept-both, reject-both, and asymmetric cases (local-only and prefix-only). Local and prefix acceptance rates are also reported.

6.3 Entropy Identifies Uncertainty in Grammatical Transitions

We observe a strong and consistent relationship between state entropy and classification error. When states are grouped by entropy and weighted by their empirical frequency in the dataset, classification error increases with entropy in the low-to-moderate range. This pattern is visible across AUX, NOUN, and DET under both local and prefix-conditioned settings, indicating that prediction failures are systematically associated with regions of higher grammatical uncertainty (Figure 3). A complementary outcome-level analysis shows that false-positive and false-negative states exhibit higher entropy than correctly classified states (Appendix C), consistent with this trend.

This relationship defines uncertainty zones within the state space: regions characterized by high entropy and elevated error rates. These zones correspond to branching points in the underlying structure.

The binary classification results further support this interpretation (Appendix D). Across targets, models exhibit high recall but lower precision. This indicates that the models successfully capture the majority of valid continuations but tend to overpredict the target class in contexts where alternative continuations are also likely. This behavior reflects the inherently multi-modal nature of high-entropy states, where the classifier must commit to a binary decision despite multiple valid next-POS possibilities.

At very high entropy levels, the relationship becomes noisier due to sparsely supported states, but the overall trend remains clear: classification difficulty increases with grammatical uncertainty.

Taken together, these findings establish entropy as a reliable indicator of intrinsic grammatical uncertainty. Low-entropy regions correspond to near-deterministic transitions that are consistently

captured by the classification models, while high-entropy regions mark the limits of deterministic prediction, where ambiguity in the underlying grammar drives error.

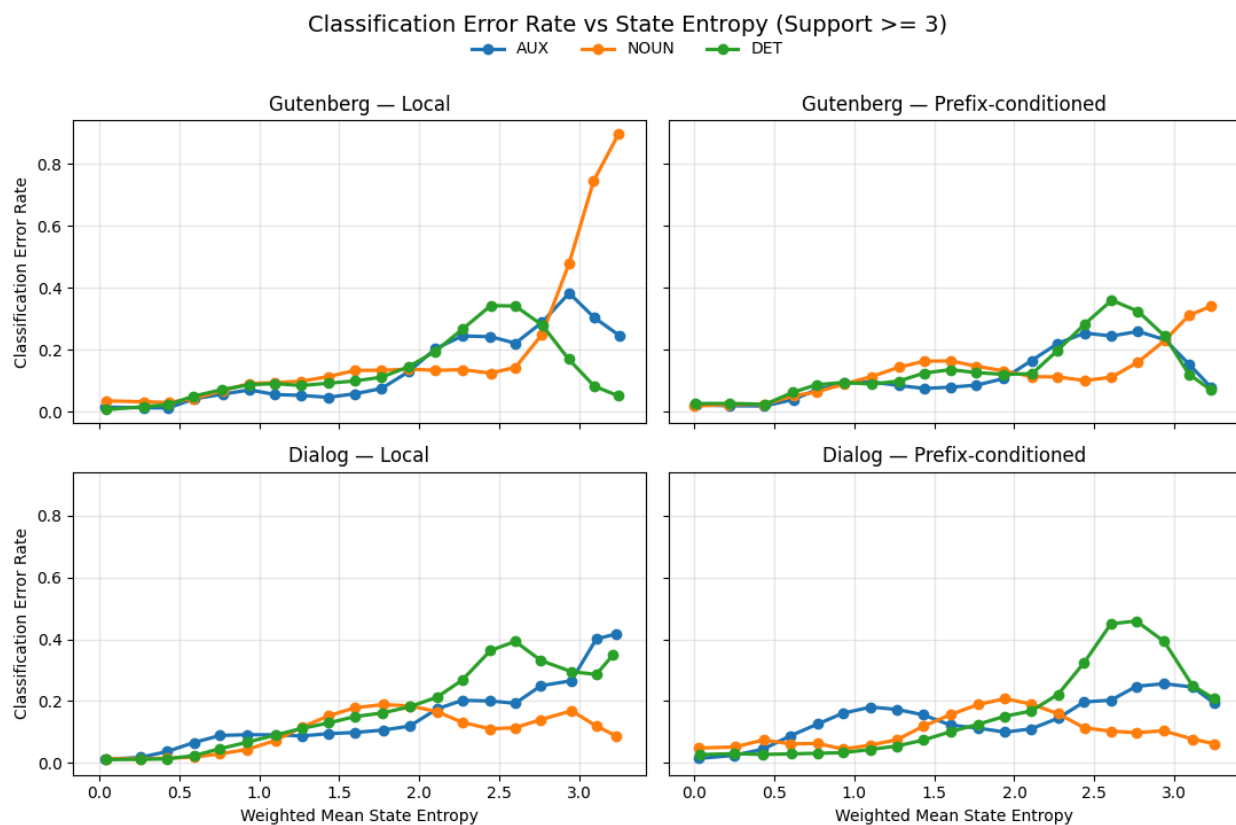


Figure 4: Classification error rate as a function of weighted mean state entropy across targets (AUX, NOUN, DET), datasets (Gutenberg, Dialog), and prompting regimes (local, prefix-conditioned). We restrict analysis to states with sufficient dataset support (≥ 3). Classification error increases with entropy in the low-to-moderate range, indicating that prediction failures concentrate in high-uncertainty regions of the grammatical state space. Variability at extreme entropy values reflects sparsely observed states.

6.4 Uncertainty Zones Drive DFA Disagreement

Across datasets and targets, agreement is highest in low-entropy regions and generally decreases as entropy increases. This pattern is most pronounced in the dialog dataset, where both agreement rate and Cohen’s kappa show a consistent downward trend across entropy buckets for all targets. These results indicate that grammatical transitions become increasingly context-sensitive as uncertainty increases, leading to divergence between local and prefix-conditioned representations.

In the Gutenberg dataset, the effect is more muted. Agreement remains high across all entropy levels, with only slight decreases observed for AUX and DET. NOUN exhibits a small increase in agreement with entropy, though the magnitude is minor. Overall, the Gutenberg results suggest that written text maintains a relatively stable grammatical structure even in higher-entropy regions, with only modest sensitivity to long-range context.

The contrast between datasets highlights an important distinction: while entropy consistently identifies regions of grammatical ambiguity, the extent to which this ambiguity affects structural consistency depends on the nature of the language. In conversational text, higher entropy corresponds to stronger divergence between context regimes, whereas in written text, grammatical structure remains comparatively robust.

Taken together, these findings show that disagreement between DFAs is not random but concentrated in entropy-defined regions of the state space. High-entropy states mark areas where local context is insufficient to fully determine grammatical transitions and where incorporating longer-range context meaningfully alters model behavior.

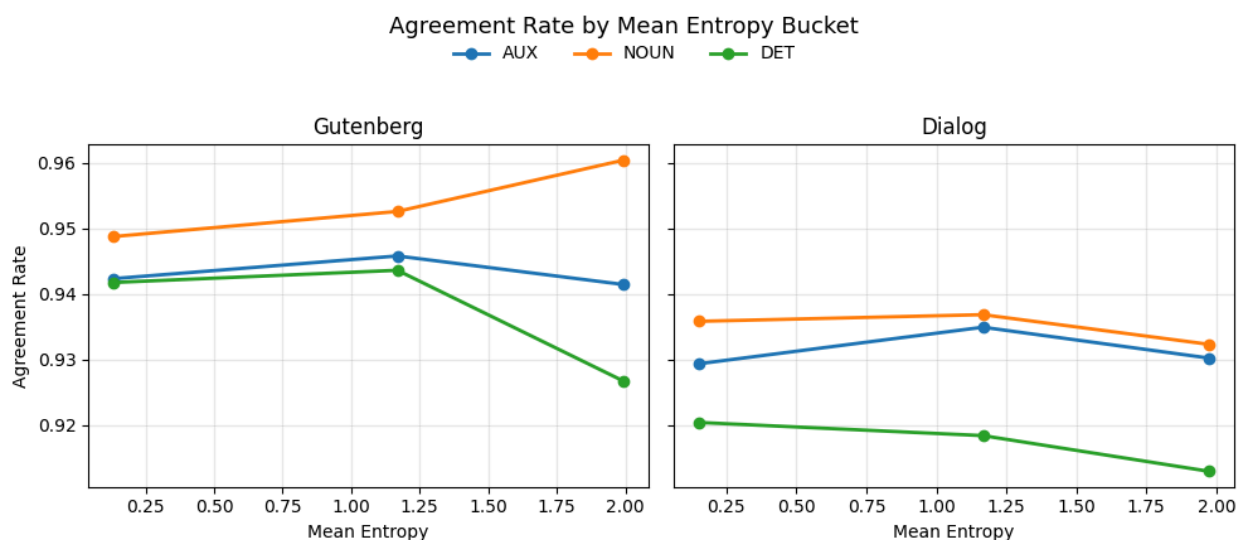


Figure 5: Agreement between local and prefix-conditioned DFAs as a function of entropy across targets (AUX, NOUN, DET) for Gutenberg (left) and dialog (right) datasets. Agreement is highest in low-entropy regions and generally decreases with increasing entropy, with a more pronounced decline observed in dialog data. This indicates that grammatical transitions become increasingly context-sensitive in high-entropy regions, particularly in conversational language.

6.5 Grammatical Structure is Robust to Stochastic Decoding

Following the temperature sweep experiment procedure outlined in Section 5.7, we examined how next-POS distributions change as temperature increases. As expected, higher temperatures introduce greater variability in token generation. We observed a monotonic increase in entropy, with mean entropy rising from 0.92 at $T = 0$ to 1.76 at $T = 0.8$, indicating a broader distribution over possible contributions. Correspondingly, divergence from the deterministic baseline increases, with mean JS divergence growing from 0.04 at $T = 0.2$ to 0.18 at $T = 0.8$.

Despite these distributional shifts, the modal structure of transitions remains largely intact. The modal match rate with the $T = 0$ distribution remains relatively high, decreasing from 0.87 at $T = 0.2$ to 0.69 at $T = 0.8$, while weighted values remain above 0.80 across all temperatures. This indicates that while lower-probability alternatives become more likely under stochastic decoding, the most probable grammatical transitions are largely preserved.

When examining the impact of temperature on DFA structure, we found that the structure remains consistent across all targets. The size of the largest SCC changes only modestly, and the fraction of states in the largest SCC increases slightly with temperature. This suggests that the core structural backbone of the automaton remains stable even as transition probabilities shift.

At the same time, acceptance behavior becomes more selective as temperature increases. The number and fraction of accepting states decrease steadily across temperatures for most targets. For instance, the acceptance fraction for AUX decreases from 9.7% at $T = 0$ to 3.8% at $T = 0.8$, with similar declines observed for DET and CCONJ. This reflects the increasing dispersion of probability mass across alternative continuations, reducing the likelihood that any single class exceeds the acceptance threshold.

Flow-based metrics further reinforce this pattern. Visitation entropy within accepting states decreases with temperature, while Gini coefficients remain relatively stable or decline slightly, indicating that visitation becomes more diffuse and less concentrated. At the same time, the proportion of visitation within the largest SCC increases slightly, suggesting that stochastic decoding encourages broader exploration within the core structural region of the DFA.

These results demonstrate that stochastic decoding primarily affects the distribution over grammatical transitions rather than the underlying structure itself. While higher temperatures introduce greater uncertainty and reduce acceptance rates, the core topology of the DFA remains largely unchanged. This indicates that the finite-state structure uncovered in earlier sections is an intrinsic property of the model’s learned grammar, rather than an artifact of greedy decoding.

TEMP	DISTRIBUTIONAL EFFECTS					STRUCTURAL EFFECTS		
	WEIGHTED ENTROPY	Δ ENTROPY	WEIGHTED JS	WEIGHTED MATCH	Δ MATCH	LARGEST SCC FRACTION	ACCEPT FRACTION	VISITATION ENTROPY
0.0	1.806	0.000	0.000	1.000	0.000	0.657	0.178	7.259
0.2	1.898	0.092	0.014	0.939	-0.061	0.677	0.154	7.094
0.5	2.114	0.308	0.045	0.882	-0.118	0.706	0.122	6.553
0.8	2.362	0.556	0.086	0.800	-0.200	0.721	0.104	6.219

Table 3: Distributional and structural effects of temperature on grammatical transitions. As temperature increases, entropy and Jensen–Shannon divergence rise, while modal agreement with the deterministic baseline decreases. Despite these changes, structural properties of the corresponding DFAs remain stable: the fraction of states in the largest SCC increases slightly, while the fraction of accepting states decreases, reflecting increased selectivity. Visitation entropy declines modestly, suggesting more diffuse traversal within the core structure.

6.6 Structural Properties of DFAs

The minimized DFAs exhibit a highly consistent pattern across all datasets, targets, and prompting regimes. Each DFA is dominated by:

- a single large SCC containing the overwhelming majority of states
- a collection of trivial SCCs, primarily singletons
- occasionally, a single small non-trivial SCC (typically of size ≈ 2)

This structure indicates that the DFA forms a single dominant recurrent core with peripheral states attached to it. The large SCC further suggests that most POS contexts participate in a shared, highly interconnected grammatical backbone rather than forming multiple independent substructures.

Analysis of the empirical support of states (i.e., how frequently they are observed in the datasets) reveals a strong asymmetry:

- states in the largest SCC receive the vast majority of support
- singleton and small SCC states have orders-of-magnitude lower support

Thus, the large SCC is not only structurally dominant but also empirically central. It captures the bulk of observed grammatical transitions during generation. Peripheral states, while numerous, contribute minimally to the overall dynamics.

Acceptance scores across all DFAs exhibit a heavily skewed and partially bimodal distribution (Appendix E). Most states are either non-accepting or weakly accepting, while a smaller subset of states form a distinct high-confidence acceptance region. This pattern is especially pronounced for NOUN and DET, where the state space separates into

- a dense cluster of low-probability contexts
- a compact set of highly predictable contexts, and
- a relatively sparse intermediate region

This separation suggests that, for these targets, the DFA effectively partitions the state space into clearly identifiable accepting and rejecting regions, with ambiguity concentrated in a limited subset of states.

Stratifying acceptance scores by SCC type reveals additional structure:

- The largest SCC contains the full spectrum of behaviors, dominated by low-score states but including high-confidence, accepting regions.
- Singleton SCCs exhibit high variability, often containing extreme values (near-zero or near-one), reflecting unstable or isolated contexts.
- Small SCCs, when present, tend to have more concentrated intermediate-to-high acceptance scores

This indicates that stable grammatical behavior is embedded within the large SCC, while peripheral SCCs capture edge cases or low-frequency transitions.

When comparing local and prefixed-conditioned DFAs, we find that their global structure remains nearly identical. The SCC decomposition is preserved, and differences in acceptance distributions are modest. However, prefix conditioning introduces systematic shifts in acceptance behavior for certain targets.

For NOUN, both datasets show a consistent shift toward higher acceptance scores under prefix conditioning, suggesting that longer context improves the predictability of noun continuations. For CCONJ (in the dialog dataset), the effect is more pronounced at the structural level, with the fraction of accepting states increasing substantially from 0.09 to 0.18. This reflects the increasing importance of discourse-level context for coordinating conjunctions.

Overall, prefix conditioning does not alter the underlying topology of the DFA but instead acts as a smooth perturbation over local predictions, selectively enhancing predictability for context-sensitive grammatical categories.

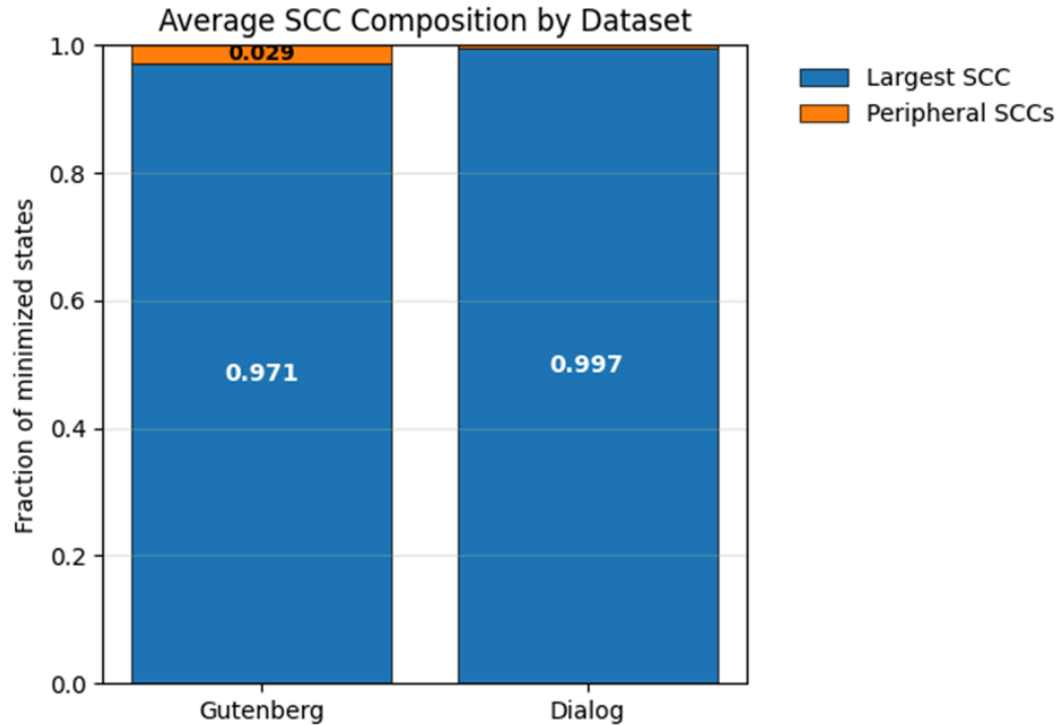


Figure 6: Average SCC composition across datasets and prompting regimes. Over 97–99% of states belong to a single dominant SCC, with the remaining states forming small peripheral components. This indicates that the DFA structure is overwhelmingly concentrated in a shared recurrent core.

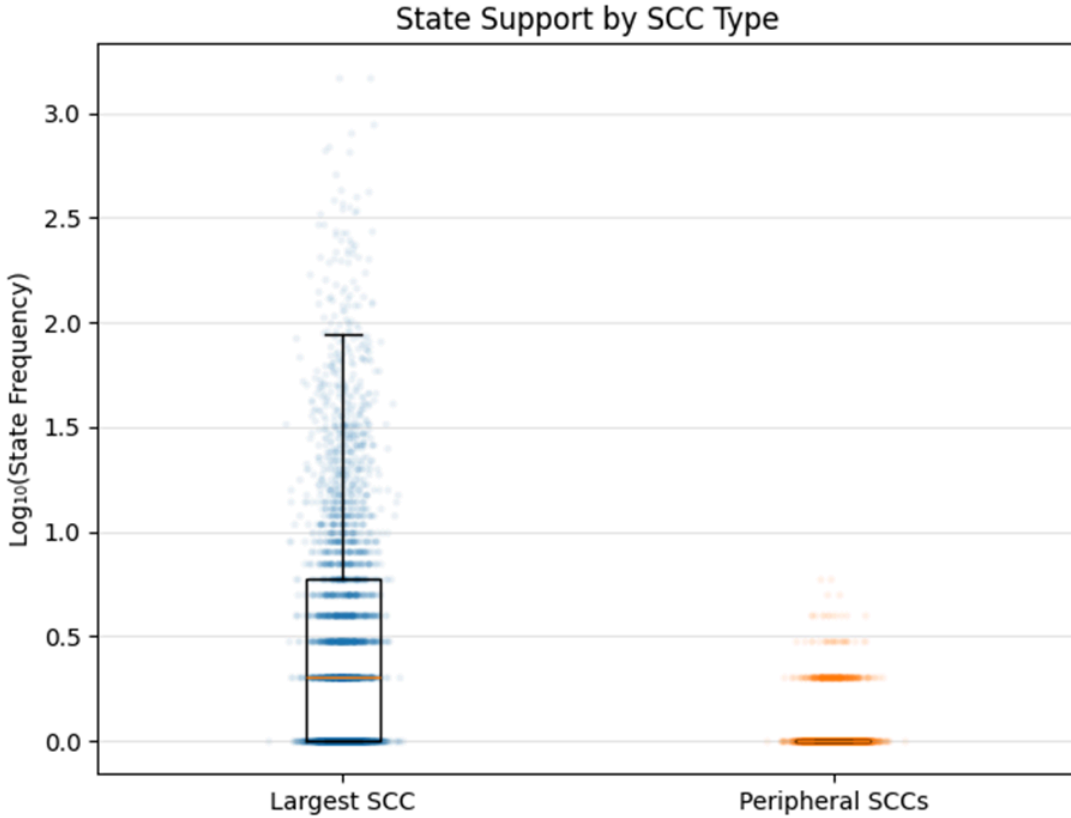


Figure 7: Distribution of log state frequency by SCC type, aggregated across datasets, targets, and prompting regimes. States within the largest SCC exhibit orders-of-magnitude higher empirical support than peripheral SCCs. This indicates that the recurrent core is not only structurally dominant but also captures the majority of observed grammatical transitions.

6.7 Flow Analysis Reveals Emergent Absorbing Behavior

Across all DFAs, traversal behavior is governed by a simple but highly structured dynamic. When initialized from arbitrary states, trajectories overwhelmingly begin within the largest SCC, reflecting its dominance in both structural connectivity and empirical support. From this core region, trajectories typically follow one of the two patterns: they either transition directly into a singleton reject state after a small number of intermediate steps, or they briefly pass through the accepting states before returning to the core and subsequently entering the reject state.

A key structural feature underlying this behavior is the presence of a singleton reject state in every DFA. This state has a self-loop, causing any trajectory that enters it to remain there indefinitely. As a result, it functions as an absorbing configuration, and random walk experiments consistently converge towards it. While initialization affects transient behavior, it does not alter this outcome. Starting from accepting states increases both the number of states visited and the number of steps before absorption, but convergence remains inevitable under random (unrestricted) exploration.

Accepting states play a distinctly transient role within this system. Across experiments, approximately 40–80% of trajectories visit accepting states, typically within the first transition step, while transitions into the reject singleton follow shortly afterward. This indicates that the DFA reliably identifies locally valid grammatical continuations, but these regions do not sustain long-term flow under stochastic traversal policies.

To better understand the origin of the absorbing behavior, we analyzed the incoming transitions into the singleton reject state. These transitions are not driven by rare or anomalous contexts. Instead, the vast majority originate from the largest SCC and are supported by high-frequency POS patterns. This suggests that the absorbing behavior emerges naturally from unrestricted exploration of ordinary grammatical transitions rather than from isolated edge cases or data sparsity.

At the same time, inspection of the underlying (pre-minimization) states that map into the reject singleton reveals that many correspond to structurally atypical or weakly supported continuations. Under DFA minimization, these states become indistinguishable with respect to future target behavior and therefore collapse into a single reject-equivalent future. The absorbing configuration thus appears to represent a loss of predictive structure relative to the target class rather than a breakdown of grammatical validity itself.

INITIALIZATION	% TRAJECTORIES VISITING ACCEPT	% TRAJECTORIES VISITING LARGEST SCC	MEAN STEP TO FIRST ACCEPT	MEAN STEPS TO ABSORPTION	MEAN STEPS (ACCEPT → ABSORPTION)	MEAN UNIQUE STATES VISITED	ABSORPTION RATE
All States (Uniform Initialization)	43.46	97.05	1.09	2.96	3.16	3.89	1.00
Accepting States	72.56	98.64	0.62	3.66	3.32	4.54	1.00
High-Confidence Accept States	76.56	96.03	0.28	2.52	2.89	3.83	1.00

Table 4: Aggregated traversal statistics across all DFAs, datasets, and prompting regimes. Trajectories typically begin in the largest SCC and frequently reach accepting states but inevitably transition into a singleton absorbing reject state. Initialization in accepting states increases exploration.

To determine whether this absorbing behavior reflects a genuine property of language or merely a consequence of the traversal process, we compare trajectories induced by natural text with stochastic random walks. Natural trajectories are generated by advancing rolling POS k-gram windows through POS-tagged Gutenberg sentences and following the corresponding DFA transitions. Random walk trajectories are initialized from the same starting states but evolve by sampling POS symbols independently from the empirical POS frequency distribution of the corpus. Unlike natural trajectories, these walks remove sequential grammatical dependencies while preserving overall POS statistics. For both regimes, we measure absorption frequency, steps to absorption, and entropy statistics. For natural traversals specifically, we additionally measure recovery behavior after entering the absorption state to assess how trajectories evolve under realistic language dynamics.

The two regimes exhibit fundamentally different dynamics. Random walks rapidly collapse into the absorbing state within only a few steps, with approximately 74% of trajectories reaching the sink within

the first five transitions. In contrast, natural language trajectories require hundreds of steps on average before first entering the absorbing configuration and almost never do so early in traversal.

More importantly, absorption under natural text is not persistent. Although natural trajectories occasionally intersect the minimized absorbing configuration, recovery occurs almost immediately in all observed cases, typically within a single transition step.

Taken together, these results show that the absorbing configuration is not a stable attractor under natural language dynamics but rather a consequence of unconstrained exploration of the state space. Randomized traversal policies rapidly diffuse into low-structure regions of the DFA, whereas natural language trajectories remain concentrated within stable, high-support grammatical manifolds. This distinction reveals an important asymmetry between stochastic exploration and real language generation: while the DFA contains absorbing regions under unrestricted traversal, natural text predominantly occupies structured low-entropy pathways that avoid long-term collapse into these configurations.

TRAJECTORY TYPE	ABSORPTION RATE	MEAN STEPS TO SINK	% ABSORBED WITHIN FIRST 5 STEPS	MEAN STATE ENTROPY OBSERVED	MEAN STEPS TO RECOVERY	RECOVERY RATE GIVEN ABSORPTION	MEAN ABSORPTION EPISODES
Natural	0.98	963.24	0.01	1.25	1.03	1.00	68.77
Empirical Random Walk	0.92	3.90	0.74	0.99	--	--	--

Table 5: Comparison of traversal dynamics under natural language trajectories and empirical random walks. Empirical random walks rapidly collapse into the absorbing state, while natural language trajectories require hundreds of steps on average before first absorption and almost never enter the sink early in traversal.

7. Discussion

7.1 A Hybrid Inference Strategy Based on Finite-State Control

Across datasets, targets, and conditioning regimes, grammatical transitions are organized around a shared, highly connected core, within which behavior ranges from near-deterministic to highly uncertain. At the same time, traversal dynamics exhibit a strong directional bias toward a terminal absorbing state, indicating that grammatical evolution is not only structured but also dynamically constrained. These findings suggest that grammatical generation in LLMs can be decomposed into low-entropy regions, where transitions are stable and highly predictable, and high-entropy regions, where multiple continuations are plausible and uncertainty is intrinsic.

This decomposition provides a natural basis for a hybrid inference strategy that combines finite-state control with LLM generation. At each generation step, the current POS context is mapped to a DFA state, and the entropy in the next POS distribution could be used to determine whether the system is operating in a low-or high-uncertainty region. In low-entropy states, generation can be delegated to a lightweight structural proxy that resolves predictable transitions without requiring full neural inference.

In high-entropy states, the LLM remains responsible for resolving ambiguity and guiding semantic continuation.

To evaluate the feasibility of this idea, we conduct a minimal routing experiment using entropy-guided model selection. Low-entropy states are routed to a lightweight n-gram baseline, while high-entropy states are routed to a GRU classifier. Despite the simplicity of the setup, the results reveal a clear tradeoff between computational cost and predictive performance. Routing approximately 25% of tokens to the cheaper model produces only a minor reduction in macro-F1 relative to the GRU-only configuration while substantially reducing the proportion of expensive inference steps. As the routing threshold increases, performance degrades gradually rather than catastrophically, indicating that much of the predictive structure resides within low-entropy regions where lightweight models remain effective.

This experiment is not intended as a production inference architecture but rather as a proof of principle demonstrating that entropy provides a meaningful signal for adaptive computation. The use of relatively simple models isolates the routing mechanism itself, showing that structural uncertainty can be used to allocate computational effort selectively rather than uniformly across all tokens.

SETTING	LOCAL			PREFIX		
	CHEAP ROUTE %	MACRO-F1	Δ F1 vs GRU	CHEAP ROUTE %	MACRO-F1	Δ F1 vs GRU
GRU only	0.000	0.364	0.000	0.000	0.339	0.000
τ 25	24.992	0.340	-0.024	25.003	0.325	-0.014
τ 50	49.995	0.339	-0.025	49.906	0.313	-0.027
τ 75	74.992	0.338	-0.026	74.992	0.308	-0.031
Baseline only	100.000	0.338	-0.026	100.000	0.303	-0.036

Table 6: Results of the entropy-guided hybrid routing experiment. Low-entropy states are routed to a lightweight n-gram baseline, while high-entropy states are routed to a GRU classifier. Routing thresholds (τ 25, τ 50, τ 75) correspond to entropy percentile cutoffs computed from state entropy distribution.

7.2 Limitations and Future Work

While the empirical findings provide strong evidence for structured grammatical dynamics, one important limitation remains. The current approach models grammatical transitions using fixed-length POS contexts (k-grams), focusing on local, short-horizon dependences. While this is sufficient to capture many phrase-level grammatical patterns, it does not fully account for long-range dependencies across clauses and sentences, discourse-level structure, and semantic constraints that influence grammatical choices. Even in the prefix-conditioned setting, the DFA operates over a compressed local representation of context, which limits its ability to model global structure.

To address this limitation, future work will extend the framework toward semantic state machines, which map higher-level contextual representations to grammatical transition patterns. This would enable the

model to capture both local grammatical structure and long-range dependencies within a unified framework.

Conclusion

In this work, we investigated grammatical structure in autoregressive text generation through the lens of finite-state representations. By constructing deterministic finite automata (DFAs) over part-of-speech (POS) transitions induced by LLM outputs, we developed a framework for analyzing how grammatical patterns evolve and how uncertainty is distributed throughout the model’s state space.

Our results reveal a consistent and interpretable grammatical organization across datasets, targets, and conditioning regimes. The extracted DFAs exhibit a dominant strongly connected component that forms a shared grammatical backbone, while entropy analysis shows that uncertainty is highly unevenly distributed. Low-entropy regions correspond to stable, near-deterministic transitions, whereas high-entropy regions correspond to intrinsically ambiguous contexts with elevated prediction error rates.

Flow analysis further demonstrates that traversal behavior depends strongly on the exploration regime. Under stochastic random walks, trajectories rapidly collapse into absorbing-reject configurations. In contrast, natural language trajectories remain concentrated within stable grammatical regions and only transiently enter absorbing states before immediately recovering. This indicates that the observed absorbing behavior is not a property of language itself but rather a consequence of unconstrained exploration of the DFA state space.

Finally, we evaluated a minimal hybrid inference strategy based on entropy-guided routing. By delegating low-entropy states to a lightweight statistical model and reserving neural inference for high-entropy regions, we show that a meaningful fraction of token predictions can bypass expensive inference with only modest performance degradation. Together, these findings suggest that LLM generation is organized around stable structural pathways interleaved with localized regions of genuine uncertainty, motivating future hybrid systems that combine finite-state control with neural computation for more efficient and interpretable language generation.

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Appendix A: spaCy Part-of-Speech Tag Labels

ADJ: Adjective

ADP: Adposition

ADV: Adverb

AUX: Auxiliary

CONJ: Conjunction

CCONJ: Coordinating conjunction

DET: Determiner

INTJ: Interjection

NOUN: Noun

NUM: Numeral

PART: Particle

PRON: Pronoun

PUNCT: Punctuation
 SCONJ: Subordinating conjunction
 SPACE: Space
 SYM: Symbol
 VERB: Verb
 X: Other

Appendix B: Complete GRU vs. Baseline Table

		GUTENBERG				DIALOG			
		LOCAL		PREFIX		LOCAL		PREFIX	
POS	MODEL	RECALL	F1	RECALL	F1	RECALL	F1	RECALL	F1
ADJ	Baseline	0.177	0.226	0.120	0.174	0.145	0.212	0.115	0.173
	GRU	0.208	0.264	0.112	0.173	0.154	0.228	0.089	0.147
ADP	Baseline	0.440	0.416	0.435	0.411	0.375	0.383	0.342	0.349
	GRU	0.516	0.458	0.497	0.440	0.405	0.401	0.358	0.359
ADV	Baseline	0.047	0.075	0.035	0.056	0.078	0.124	0.055	0.091
	GRU	0.049	0.090	0.015	0.028	0.100	0.159	0.031	0.057
AUX	Baseline	0.564	0.559	0.499	0.509	0.643	0.658	0.596	0.603
	GRU	0.584	0.576	0.537	0.537	0.666	0.679	0.624	0.618
CCONJ	Baseline	0.594	0.541	0.505	0.433	0.775	0.721	0.601	0.568
	GRU	0.701	0.568	0.587	0.480	0.779	0.734	0.648	0.588
DET	Baseline	0.669	0.641	0.589	0.566	0.674	0.594	0.596	0.539
	GRU	0.695	0.646	0.598	0.580	0.685	0.593	0.636	0.561
INTJ	Baseline	0.100	0.167	0.000	0.000	0.231	0.326	0.111	0.159
	GRU	0.500	0.625	0.000	0.000	0.238	0.319	0.158	0.241

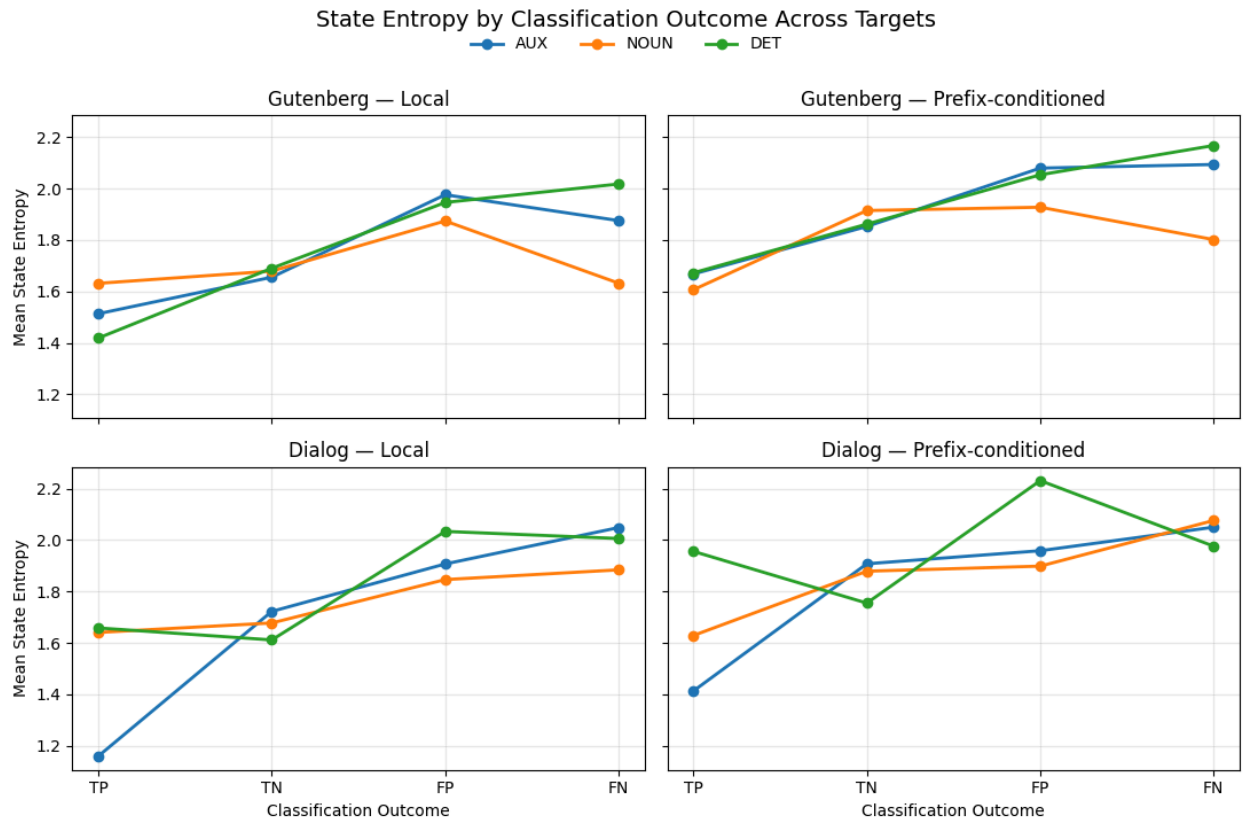
NOUN	Baseline	0.650	0.614	0.678	0.628	0.609	0.589	0.612	0.581
	GRU	0.694	0.626	0.740	0.640	0.643	0.607	0.661	0.614
NUM	Baseline	0.152	0.215	0.204	0.271	0.126	0.199	0.125	0.203
	GRU	0.271	0.373	0.166	0.227	0.221	0.331	0.205	0.300
PART	Baseline	0.130	0.178	0.127	0.167	0.275	0.313	0.139	0.187
	GRU	0.101	0.155	0.106	0.150	0.237	0.279	0.136	0.189
PRON	Baseline	0.290	0.361	0.299	0.344	0.409	0.468	0.474	0.494
	GRU	0.306	0.387	0.319	0.362	0.441	0.489	0.507	0.531
PROPN	Baseline	0.178	0.243	0.169	0.217	0.097	0.147	0.078	0.122
	GRU	0.125	0.187	0.152	0.212	0.083	0.140	0.036	0.065
PUNCT	Baseline	0.594	0.522	0.582	0.484	0.629	0.539	0.623	0.530
	GRU	0.565	0.552	0.547	0.514	0.634	0.577	0.614	0.550
SCONJ	Baseline	0.013	0.024	0.015	0.026	0.046	0.076	0.040	0.065
	GRU	0.005	0.009	0.000	0.000	0.038	0.068	0.026	0.048
SYM	Baseline	0.136	0.214	0.154	0.211	0.095	0.138	0.067	0.095
	GRU	0.000	0.000	0.125	0.222	0.046	0.077	0.000	0.000
VERB	Baseline	0.275	0.324	0.302	0.343	0.381	0.399	0.438	0.419
	GRU	0.325	0.360	0.323	0.359	0.412	0.419	0.482	0.449
X	Baseline	0.324	0.421	0.433	0.520	0.000	0.000	0.000	0.000
	GRU	0.467	0.519	0.484	0.526	0.250	0.400	0.000	0.000

Appendix C: Binary Classification Performance

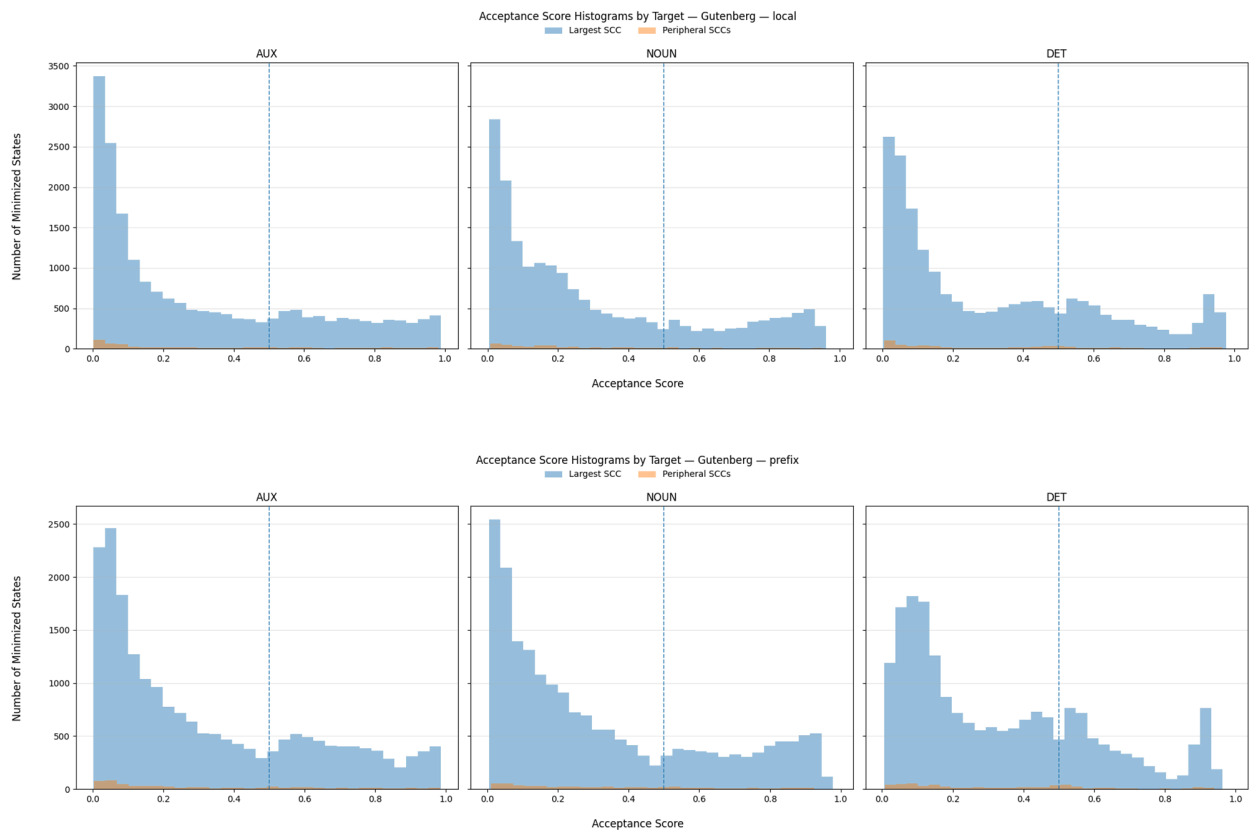
	GUTENBERG	DIALOG
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POS	MODEL	PRECISION	RECALL	F1	PRECISION	RECALL	F1
AUX	Local	0.381	0.878	0.532	0.469	0.880	0.612
	Prefix	0.338	0.818	0.478	0.406	0.820	0.543
NOUN	Local	0.465	0.805	0.590	0.421	0.816	0.555
	Prefix	0.489	0.817	0.612	0.429	0.789	0.555
DET	Local	0.502	0.826	0.624	0.442	0.835	0.578
	Prefix	0.429	0.774	0.552	0.367	0.835	0.510
CCONJ	Local	--	--	--	0.536	0.900	0.672
	Prefix	--	--	--	0.311	0.839	0.454

Appendix D: State Entropy by Classification Outcome

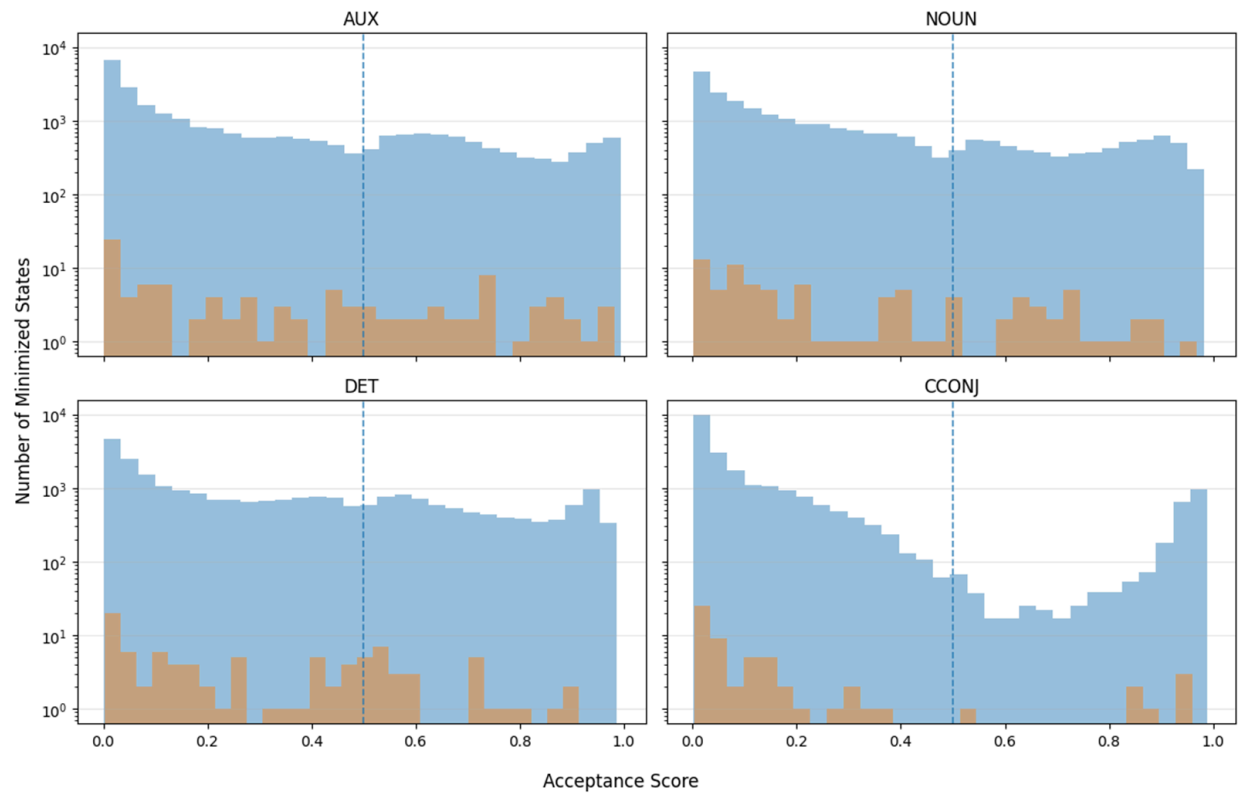


Appendix E: Acceptance Score Histograms By Target



Acceptance Score Histograms by Target — Dialog — local — (Log Scale)

■ Largest SCC ■ Peripheral SCCs



Acceptance Score Histograms by Target — Dialog — prefix — (Log Scale)

■ Largest SCC ■ Peripheral SCCs

